CARDIO-CODE

DSP-Driven Arrhythmia Detection in Motion-Corrupted PPG

Signal processing only, no machine learning

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Overview

1. Problem
2. Our Pipeline
3. Testing and Results
4. Contribution and Limitations

The Problem

A heartbeat hidden in noise

A wearable story

- A wrist band records a few seconds of irregular heartbeat during a walk.
- Arm swing adds pulse-like waves; sweat makes the signal drift.
- The irregular beats are hidden, and **nothing flags them**.

Our goal

The software layer between a noisy signal and a **trustworthy, flagged insight**.

Why it is hard

- Arm motion (about 1–2 Hz) sits in the **same frequency range** as the pulse.
- A fixed filter cannot separate them.
- Irregular rhythm must not be mistaken for noise, and noise must not be mistaken for irregular rhythm.

Task, Output and Constraints

Output: `predictions.csv`, one row per input sample

`Timestamp`	time of the sample (s)
`Is_peak`	1 if a heartbeat peak is here
`Ibi_ms`	time between the last two beats (ms)
`ano_flag`	1 inside a suspected abnormal rhythm
`confidence_score`	0–1: how far this part can be trusted

Input

CSV with a PPG *(optical pulse)* or ECG *(electrical)* column; optional time and accelerometer columns.

Constraints

- Time-domain or short-time spectral processing only.
- No ML, neural networks or pretrained models.
- Every step explainable.

Scoring criteria

- Signal accuracy
- Anomaly detection
- Computational efficiency

The Data We Were Given

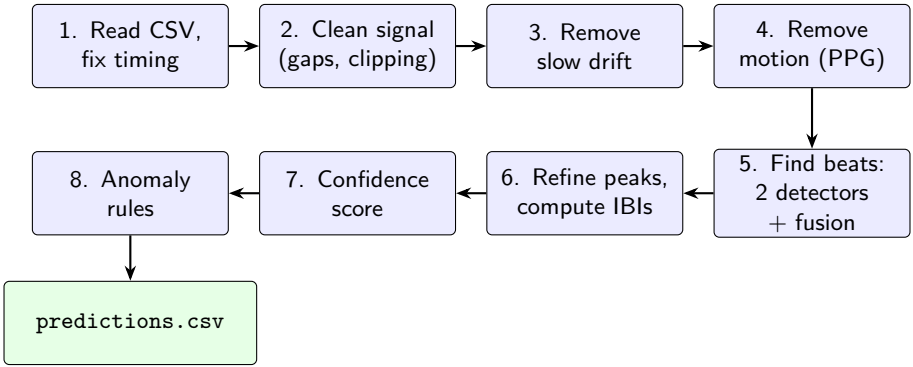
ecg.csv: MIT-BIH record 231

- 650 000 samples at 360 Hz (30.1 min), two ECG leads.
- Contains a **2:1 AV heart block**: every second beat is dropped.
  - Normal rhythm: IBI $\approx$ 961 ms (62 bpm).
  - Block: IBI $\approx$ 1657 ms (36 bpm).

Two consequences for our design

1. The anomaly is **slow and regular**, not AF, so we need several independent rules, not just one irregularity test.
2. The hidden test is **motion-corrupted PPG**, but our data is clean ECG, so we built our own stress tests with known answers, and the program picks ECG or PPG processing automatically.

Pipeline Overview



ECG and PPG share this flow and differ in steps 3–5: an ECG beat is a sharp spike, a PPG beat is a smooth hump. The mode is chosen from the column name, or from the frequency content.

Steps 1–3: Input, Cleaning and Drift

Input handling

- The delimiter, column roles (by name) and time unit are detected automatically.
- Jittery or out-of-order timestamps are resampled onto an even grid.
- NaN runs and gaps longer than 1 s are invalid: no beats, confidence 0.

Baseline wander (slow drift)

- **Butterworth filter** *(a standard, smooth frequency sieve)*, run **zero-phase**: forwards, then backwards, so beats do not shift in time.
- ECG: 0.5 Hz high-pass. PPG: 0.5–8 Hz band-pass.
- ECG only: adaptive 50/60 Hz **notch** *(removes mains hum)*, used only if the hum is >10 dB above its surroundings.

Step 4: Motion Artifact in PPG

Two paths

With accelerometer columns

NLMS adaptive filter *(keeps re-tuning itself to predict how arm motion leaks into the PPG, then subtracts it)*

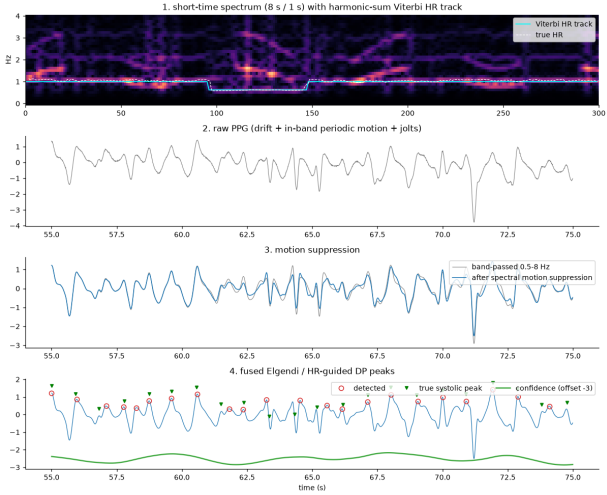
- 8 taps per axis, step size 0.01, runs at about 25 Hz.
- Used only if PPG/accelerometer **coherence** *(how strongly they move together)* in 0.5–5 Hz exceeds 0.1.

Without an accelerometer

Short-time spectrum (8 s Hann window, 1 s step, at about 25 Hz)

- **Harmonic sum** *(add the 2×/3× overtones back)* makes the true heart rate stand out.
- **Viterbi tracker** *(the smoothest strong path through time)* follows the heart rate.
- Strong lines more than 0.15 Hz from every heart-rate harmonic are motion: re-synthesised and subtracted.

Motion Handling on Synthetic PPG



1. Spectrogram with the Viterbi heart-rate track (cyan) against the true heart rate.
2. Raw PPG: drift, in-band motion and jolts.
3. Before and after spectral motion removal.
4. Fused beat detections and the confidence score.

Synthetic test signal, motion level 0.5.

Step 5: Beat Detection with Two Detectors

A. Threshold detector

- **ECG, Pan–Tompkins:** 5–15 Hz band → derivative → square → 150 ms moving average → adaptive threshold; 250 ms refractory *(no two beats closer)*; search-back for missed beats.
- **PPG, Elgendi:** 111 ms and 667 ms moving averages, local 10 s offset, 300 ms refractory.

B. Heart-rate-guided selector

Dynamic programming *(picks the best whole sequence at once)*: candidate peaks are scored by rise amplitude; the chosen sequence maximises total score minus a penalty for straying from the expected IBI (weight 3). Real dropped beats survive (tolerance 0.3).

Noise-gated fusion (per 10 s window)

1. $\geq$90 % of beats agree within 100 ms $\Rightarrow$ keep A.
2. Window is clean (noise ratio $<$ 0.45 ECG, $<$ 0.20 PPG) $\Rightarrow$ keep A, **even if the rhythm is irregular**.
3. Otherwise keep the steadier sequence, biased towards A (1.25 ECG, 2.2 PPG).

Why two?

B is robust to noise but expects a regular rhythm. Rule 2 stops it erasing real atrial fibrillation (AF).

Steps 6–7: Peak Refinement, IBI and Confidence

Peak refinement and IBI

- **Parabolic interpolation** *(a tiny parabola through the top 3 samples)* gives timing finer than one sample.
- ECG: detect whether beats point up or down, then snap each peak to its tip.
- IBI = time between consecutive peaks; IBIs outside 0.25–3 s are ignored by the rules.

Confidence score: "a real beat looks like its neighbours"

- Per beat: correlation with the median **template** of beats within 5 s $\times$ amplitude consistency $\times$ validity.
- Interpolated to every sample, smoothed over 2 s.
- Forced to 0 in gaps over 3 s, on clipped or flat samples and in invalid regions.

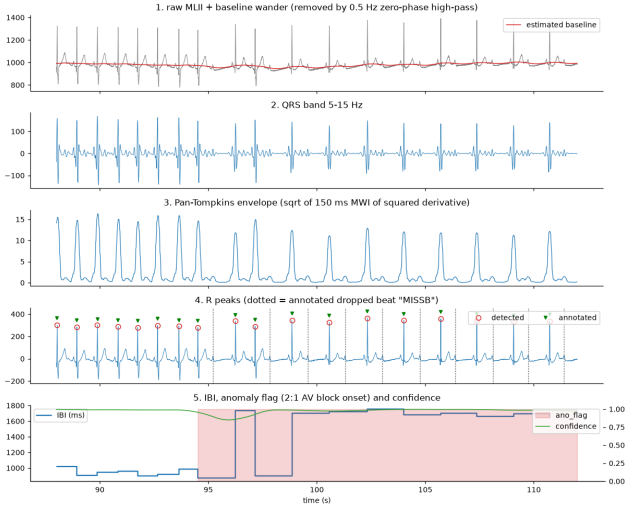
Step 8: Five Explainable Anomaly Rules

Reference IBI = median of the previous 8 valid IBIs. Rules run only when both beats have quality $\geq$ 0.5.

Rule	**Fires when**
Pause / dropped beat	IBI $>$ 1.5$\times$ reference, or IBI $>$ 2 s
Bradycardia (slow heart)	5-beat median heart rate $<$ 45 bpm
Premature beat	IBI $<$ 0.8$\times$ reference and IBI + next $>$ 1.6$\times$ reference
Ectopic shape (ECG only)	template correlation $<$ 0.8, neighbours $>$ 0.9, IBI $<$ 0.92$\times$ reference
AF-type irregularity	16-beat window: median $\lvert\Delta\text{IBI}\rvert$/median IBI $>$ 0.10 **and** nRMSSD $>$ 0.15 **and** (turning-point test **or** Poincaré SD1/SD2 $>$ 0.6)

- Flagged spans less than 6 s apart are merged.
- `--events` exports each episode with its rule and a plain-text reason.

Example: Catching the Heart Block in ecg.csv



- Panels: raw signal and drift → QRS band → Pan–Tompkins envelope → detected vs annotated peaks → IBI, flag and confidence.
- At onset the IBI jumps from about 960 to 1660 ms, which is $> 1.5\times$ the reference, so the **pause** rule fires.
- During the block the heart rate is about 36 bpm (< 45), so the **bradycardia** rule keeps it flagged.

How We Tested

Test data with known answers

- `ecg.csv` (record 231) with annotations.
- Record 231 + real NSTDB noise: electrode motion, muscle, baseline wander at 12 / 6 / 0 dB.
- Seven real MIT-BIH arrhythmia records.
- **Synthetic** PPG from real beat times: pulse about 220 ms after each beat, breathing, drift, in-band motion bursts and jolts.
- Two independent synthetic PPG fixtures, with and without an accelerometer.

Metrics

- A beat counts if within ± 150 ms of a true beat.
- Sensitivity *(true beats found)*, precision *(detections that are real)*, F1 *(their balance)*.
- IBI mean absolute error (MAE).
- Anomaly F1, per sample.

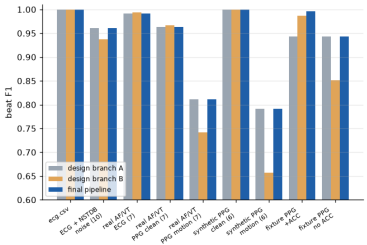
Results

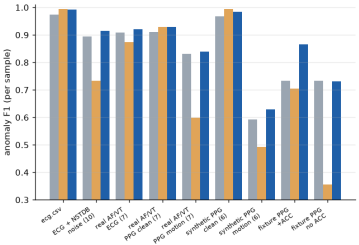
Data source	Beat F1	IBI MAE	Anomaly F1
`ecg.csv` (real clean ECG; also used for tuning)	1.000	0.99 ms	0.992
44 team-built stress sets (incl. synthetic PPG), mean	0.925	–	0.878
Synthetic PPG fixture **with** accelerometer	0.996	–	0.865
Synthetic PPG fixture **without** accelerometer	0.943	–	0.730
Worst: ECG + electrode-motion noise at 0 dB	0.776	–	0.484
Worst: synthetic PPG with strong motion	0.72–0.74	–	–

- Runtime: about 2 s for a 30-minute record (from our report).
- The confidence score drops where accuracy drops, so the output says when not to trust it.

Design by Measurement

Two full design branches, one scorer





- We built two complete designs (A and B) and scored them on identical files.
- The final pipeline keeps the better component from each.
- 44 sets, beat / anomaly F1:
  - A: 0.925 / 0.860
  - B: 0.892 / 0.774
  - **Final: 0.925 / 0.878**

What Is New

The building blocks (Butterworth, Pan–Tompkins, Elgendi, NLMS, Viterbi, template quality) are published methods. Our contribution is how they are combined:

Contributions

- **Noise-gated fusion** of a threshold detector and a heart-rate-guided selector, so irregular rhythms such as AF are not erased.
- **Spectral motion removal** that works without an accelerometer.
- **Design by measurement**: two full branches benchmarked head to head.
- **Explainable output**: every anomaly has a named rule and a reason.

Limitations

Current limits

- All PPG test data is synthetic; there is no real-PPG validation yet.
- No score on the hidden test set is available to us.
- Tuned on our own test sets, so scores may be lower on unseen data.
- Very strong in-band motion lowers accuracy.
- PPG peaks are not shifted back by the $\sim$250 ms pulse travel time.
- Offline processing; a research prototype, not a diagnostic device.
- The AF rule needs 16 beats; very short events fall to the premature-beat and pause rules.

Demo

Generate a test signal (no external data needed) and run

```
python3 dev/make_fixtures.py --out-dir demo_data
python3 pipeline.py -i demo_data/ppg_acc.csv -o demo_data/pred_acc.csv \
        --annotation demo_data/ppg_acc_ann.txt --events demo_data/events_acc.json
```

What to look at

- The summary line: sampling rate, resampled, accelerometer used, beats, heart rate.
- The accuracy line printed against the annotations.
- `events_acc.json`: each anomaly episode with its rule and reason.

Thank you

Questions?

Team Ramanujan: Darshan, Piyush